

FIXES

Building for Real With Digital Blocks

To improve community structures with citizens' input, the United Nations uses a computer game inspired by Lego.

Oct. 15, 2019

**By Tina Rosenberg**

Ms. Rosenberg is a co-founder of the Solutions Journalism Network, which supports rigorous reporting about responses to social problems.

See more of our coverage in your search results.

Add The New York Times on Google ↗

In 2011, Jorgen Hallstrom, a project manager at Swedish Building Services, was redesigning public housing complexes around Stockholm. The buildings were rundown and bleak, the public spaces uninviting.

Mr. Hallstrom, like any developer, wanted to get residents' views. But he knew that very few would come to hear about the project, and those that did would struggle to understand the architectural drawings. For the untrained, it's not obvious how a two-dimensional drawing represents three-dimensional space. And it was likely that the same voices as always would dominate the meeting.

He worried that the result would be a design that the residents wouldn't use and didn't value enough to maintain.

Mr. Hallstrom talked about this problem at home with his family. His 11-year-old son looked up from the computer game he was playing. “Why don’t you try this?” he said.

The boy was playing Minecraft, a game invented two years before by the Swedish company Mojang. It’s not a typical video game: In classic Minecraft, there’s no enemy, you don’t kill anyone or anything, and there’s nothing to win. It’s just a virtual version of Lego. Using Minecraft’s building blocks, players create castles, villages, entire worlds. Cody Sumter of the M.I.T. Media Lab said in 2012 that Minecraft’s inventor hadn’t just built a game. “He’s tricked 40 million people into learning to use a CAD program,” Mr. Sumter said, using the acronym for computer-aided design.

Forty million? More like 112 million active players every month. Today, it’s the most-downloaded video game ever, according to a spokesman for Microsoft, which bought Mojang in 2014.

Mr. Hallstrom called Mojang the next day, and with their cooperation he set up a few trial projects around Stockholm. Word spread. In February 2013, residents of the Kibera slum in Nairobi used it to redesign a sports field.

The branch of the United Nations that works to improve cities, U.N.-Habitat, brought Minecraft to Kibera. “When we first started, it felt like a crazy idea,” said Pontus Westerberg, the program management officer. “But the most surprising thing is that it actually works. We see positive engagement in these workshops. It’s a tool that gives ordinary people a way to think like an architect.”

Mr. Westerberg said that Minecraft wouldn’t be good for planning a neighborhood or designing a building — it’s too blocky and low-resolution, and there’s not enough detail. (The only lighting choice, for example, is a flaming torch.) But it’s useful for planning public spaces, he said, because you want people to stick to the big questions. Residents need to decide where to put lights. What they look like is a question for a designer, later on.

Today, people all over the world use Minecraft to design public space. The projects are a collaboration of U.N.-Habitat and the Block by Block foundation, which is funded by Microsoft and Mojang. Together they work with community groups, usually in the slums of developing countries, where there is often just a bald patch of dust or no public space at all.



Students showing designs for pedestrian safety at a workshop in Hanoi.

Katla Studio

The point is to make more than a pleasant gathering place. In Hanoi, for example, 45 schoolgirls used Minecraft to redesign pedestrian areas to create a safer walk to school. In Mitrovica, Kosovo, a city divided by a river into Albanian and Serb zones, people from both ethnicities came together to use Minecraft to improve the riverbank and market neighborhood at the bridge — the only place used by both communities. In Mdantsane, a town in South Africa's Eastern Cape, students made Minecraft improvements to their school for boys with physical disabilities. Refugees from Sudan in Kalobeyei, Kenya used Minecraft to design part of their settlement: a place for refugees and local residents to mix, with shade structures, a playground and a motorcycle taxi stand.

Block by Block accepts applications from community groups who can show they have support from the local government. The foundation chooses 10 to 20 projects to fund each year. So far, it's finished nearly 100 projects, in 35 countries. Block by Block funds a piece of the process, or all of it: the public consultation with Minecraft, further discussions to rank priorities, architects' drawings, permits and, finally, construction. Block by Block also provides its methodology free to projects it doesn't select for funding.

A Minecraft public session starts with community organizations putting up fliers and spreading the word. “The No. 1 challenge for city planners in public spaces is engaging the community,” said Deirdre Quarnstrom, who manages Microsoft’s efforts to expand use of Minecraft in education, in China, and by players with disabilities.

The opportunity to play — in some places, just the chance to use a computer — is a lure, especially for the young. Mr. Westerberg said that in the Middle East, Latin America and some parts of Asia, Minecraft is widely known; in Africa and other places in Asia, not so much.

When residents gather, they take a field trip to the site, coming back with photos and notes. They sit down in front of computers loaded with a Minecraft model of the site — ideally, one laptop for every three people.

Then they learn to play the game. It’s designed to be intuitive. “If you have basic computer skills, then two hours of training is enough,” said Luis Miguel Artieda, program director of the Avina Foundation in Lima, Peru, which manages and helps to fund Minecraft redesign projects in Latin America. Mr. Westerberg and others said that even people who have never used a computer can build in Minecraft in three or four hours.

Avina’s biggest project is in the Villa El Salvador slum of Lima, an enormous park called Parque Mamá Lucinda, built on a space that spent decades as a garbage dump. It’s named for a resident, Lucinda Terrazas, now 80, who has been fighting for such a park for 40 years. “This was certainly her first experience with computers,” Mr. Artieda said. “She sat at a computer with two or three people, so she didn’t have to move the mouse herself.”

After the teams finish their designs, they present them one by one, and the group discusses priorities. This design process can last a weekend, or stretch out over multiple weeks — six Saturdays, for example.

In a Minecraft consultation, more people show up and give their idea than in a traditional community consultation. “There are power dynamics,” Nicolette Pingo, development facilitation manager of the Johannesburg Development Agency, said of traditional consultations. “Most of the time, it’s men speaking a lot,” she said. “We don’t hear the voice of women.” Or of youth, or homeless people — who care intensely about the design of public space. “In Minecraft, even if you are shy you can put your point across.” Even if you don’t speak the dominant language.

Minecraft also helps people understand the project. An architectural drawing usually offers only a bird’s-eye view. But in Minecraft, players walk around a building, go inside, look out windows, climb a tree, run around a soccer pitch.

Some who work with Minecraft say it also encourages a richer discussion and more ideas. Many teenagers dream of playing video games for a living; James Delaney actually does it. Six years ago, when he was 17, he founded Blockworks, a company in London that conducts educational and marketing projects with Minecraft. (Here are some jaw-dropping examples; check out the three maps of London’s Great Fire of 1666).

In the undergraduate thesis he wrote last year, Mr. Delaney told the story of the Victoria and Albert Museum. When it opened a new entrance in 2017, the museum recruited Blockworks to run a Minecraft workshop asking children to reimagine the building’s dazzling white porcelain-tiled courtyard. None of the players had any verbal criticism of the courtyard. But most submitted redesigns that were very different; 75 percent of the players, for example, put in trees, water or plants.

As with any kind of public consultation, a Minecraft session is only the first of many difficult steps. U.N.-Habitat’s very first project, in Nairobi, eventually failed because different groups claimed rights to the land.

Mr. Artieda said that the Villa El Salvador project, which began in 2014, is still only partially completed. “When the local government changed in 2015, we had to go back to the beginning,” he said. “Everything was lost. We had to do a lot of work to convince the new officials. Now they are more convinced. But he said they still haven’t done what is needed to complete the park.

Mr. Artieda's group, Avina, has done 12 projects in six Latin American countries using Minecraft. "The hardest part has been to get a sustained commitment from local authorities," he said. "Projects advance when have a local government that stays in office and is committed to the idea."

New technology can make designing with Minecraft even more vivid. In Braamfontein, a neighborhood in Johannesburg, the Swedish technology company Ericsson joined a project to design a plaza, adding a mixed reality component. Each team's Minecraft redesign was uploaded onto a virtual reality version. Looking through a special phone, the young designers could walk around the plaza inside their own version of the space. (Here's a video.)

The real Minecraft challenge is to make it more than a cool one-time event, since special projects funded by outside foundations are not sustainable.

Mr. Westerberg said that many places have started formally including public participation in national urban policies and city handbooks. And he said that in Scotland, South Africa, Indonesia, Bahrain and the United States, U.N.-Habitat has trained local governments and civic organizations to do their own Minecraft-led consultations.

Mr. Artieda said that in Latin America, the projects give people unaccustomed to being listened to a taste of what they can do by working together. "It's the human connections," he said. "These projects cost little, have local leadership and people quickly benefit from their own effort. They learn that with their own tools, they can attack different problems. Public space construction — it's an excuse to construct a community."

Tina Rosenberg won a Pulitzer Prize for her book "The Haunted Land: Facing Europe's Ghosts After Communism." She is a former editorial writer for The Times and the author, most recently, of "Join the Club: How Peer Pressure Can Transform the World" and the World War II spy story e-book "D for Deception."

To receive email alerts for Fixes columns, sign up here.

The Times is committed to publishing a diversity of letters to the editor. We'd like to hear what you think about this or any of our articles. Here are some tips. And here's our email: letters@nytimes.com.

Follow The New York Times Opinion section on Facebook, Twitter (@NYTopinion) and Instagram.

